

Predicting When Collection Statistics Help Top- k Selection: From Retrieval to Attention and Expert Routing

Team ANLP Doomsday

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Abstract

Retrieval, long-context attention and mixture-of-experts routing each select a few items by an inner product against learned features and a hard top- k . Only retrieval corrects that score, in BM25, with inverse document frequency, saturation and length normalisation. We ask when the correction helps, and whether that can be predicted before running it. In the completed retrieval setting, fourteen sparse representations built from one frozen encoder are scored under all eight combinations of IDF, saturation and length normalisation, in 182 primary rows on thirteen collections and 1,226 sensitivity rows on ten, from two runs that agree to a median 0.0007 nDCG@10. The correction helps in 90% of primary rows. A Shapley decomposition assigns 61% of the gain to saturation, which needs no collection statistics, and 39% to IDF and length together, within a point on both runs. The pre-registered predictor, a ridge regression on three frequency diagnostics, does no better than the majority sign, because the effect of saturation depends on how unequal the stored values are, which the index shows, while the value of IDF and length depends on relevance, which it does not. Scoring each configuration against a reference ranking supplies that information without labels. It predicts the gain from collection statistics on unseen corpora with R^2 of 0.52 to 0.66, and using collection statistics only where it predicts a gain recovers 61% of the gap to an oracle. Five pre-registered tests of the mechanism pass. Attention and routing are implemented and are next.

1 Problem Statement

Selection by inner product. A learned sparse retriever scores a query q against a document d as $\langle u_q, u_d \rangle$ over a vocabulary of V non-negative features and returns the top k . A block-sparse attention kernel scores KV page i for query q_t as $\langle q_t, c_i \rangle$ over page summaries and attends to the top-

k pages (Tang et al., 2024). A mixture-of-experts router gates on $\langle h_t, w_e \rangle$ and sends the token to its top- k experts (Shazeer et al., 2017; Fedus et al., 2022). All three compute an inner product and keep the top k .

The correction. For query and document vectors x_q, x_d with non-negative coordinates we score

$$S(q, d) = \sum_j x_{qj} w_j T(x_{dj}, \ell_d), \quad (1)$$

where w_j is one or the smoothed inverse document frequency $\text{idf}_j = \ln(1 + (N - df_j + 0.5)/(df_j + 0.5))$, with N documents of which df_j have feature j , the length factor is $\ell_d = 1$ or $1 - b + bL_d/\bar{L}$ with $L_d = \sum_j x_{dj}$, and $T(f, \ell)$ is f/ℓ or the saturated $(k_1 + 1)f/(f + k_1\ell)$. Each of the three components is a switch, so a configuration is a bit string $c \in \{0, 1\}^3$ in the order IDF, saturation, length. Configuration 000 is the plain inner product and 111 is BM25 (Robertson and Zaragoza, 2009). IDF and length normalisation read collection statistics. Saturation reads only the document.

The unit of observation. Collection frequency belongs to a corpus, so one row of the analysis pairs a representation with a collection, at a support budget (query and document feature caps), BM25 parameters (k_1, b) and a basis seed. Each row carries eight nDCG@10 values Q_c , one per configuration.

What is predicted. The pre-registered target is the margin $m = (Q_{111} - Q_{000})/\text{sd}(Q_{000}, \dots, Q_{111})$, a standardised effect on the same scale in every setting. Section 7 shows that m is poorly shaped for prediction, and we add targets defined on the same eight values. The Shapley value ϕ_s of switch s is its marginal effect averaged over every order in which the switches can be turned on (Shapley, 1953).

Writing Q_A for the configuration with the switches in A on,

$$\phi_s = \sum_{A \not\ni s} \frac{|A|!(2-|A|)!}{6} (Q_{A \cup \{s\}} - Q_A), \quad (2)$$

and the three values sum exactly to $Q_{111} - Q_{000}$. The collection gain $G = Q_{111} - Q_{010}$ is the value of IDF and length normalisation once saturation is applied. Its sign is the decision the project is about, whether to use collection statistics for a given representation and collection. A predictor is a function of label-free quantities that returns m , ϕ_s or G for a row. It is scored by sign accuracy against the majority sign, held-out R^2 and Spearman correlation, and, for G , by the nDCG@10 obtained when collection statistics are used only where the prediction is positive.

Transfer. Eq. (1) applies to any selection with a non-negative score and a selection count. For KV page i at decode step t , the feature value is the softplus of the Quest criticality $r_{t,i}$, the collection frequency is sf_i , the fraction of calibration queries for which page i enters the top- k among the pages that exist at that point, and the length is the norm of the page’s concatenated key minima and maxima. For expert e at a token, the value is the router probability p_e , the frequency is the share of calibration tokens routed to e , and the length is the norm of the router logits. The selected experts are combined with their original gate weights, so the correction changes only which experts run. In attention a row is a model, RULER task, context length and token budget, with RULER accuracy as the metric; in routing it is a model and task, with negative token log-likelihood on WikiText and zero-shot MMLU accuracy. A predictor fitted in retrieval is applied there without refitting. It transfers if its predicted sign is correct on at least 70% of rows and its mean absolute error is at most 1.5 times its corpus-held-out RMSE in retrieval.

2 Motivation and Research Gap

The inner-product selectors of Section 1 use the raw match score, with no account of how often a feature is selected across the collection, how evidence accumulates within an item, or how large the item is. Classical lexical retrieval corrects its raw match score for all three, with parts that arrived separately, inverse document frequency (Spärck Jones, 1972), term-frequency saturation (Robertson and Walker, 1994) and document length normalisation

(Singhal et al., 1996), combined in BM25 (Robertson and Zaragoza, 2009). Inside the transformer only fragments have been reinvented. Loss-free load balancing adds a per-expert bias driven by recent load (Wang et al., 2024a), an additive frequency discount with no saturation or length term, and streaming inference keeps the most-attended early keys as an exception (Xiao et al., 2024).

Three papers have scored the same learned features both with and without the correction, all in retrieval. BM25 wins on sparse-autoencoder features of a dense retriever (Clavié et al., 2026) and on SAE concept units (Park et al., 2025), and comes out level on visual words (Han et al., 2026). Clavié et al. and Han et al. give the same explanation, that BM25 helps while a vocabulary keeps a Zipfian frequency curve. None of the three varies the vocabulary systematically, and none attributes the gain to the three components. The Zipfian account has not been turned into a quantity that can be computed before the experiment, nor checked where the same computation runs inside a model.

3 Contributions

1. A controlled grid, built and run. Fourteen sparse representations from one frozen encoder, seven bases at two granularities, are scored under all eight configurations on thirteen collections, with sensitivity families over support budget, k_1 and b , basis seed and collection-statistic controls, in 182 primary rows and 1,226 sensitivity rows from two independent runs.
2. An attribution of the gain. Shapley values over the three switches show that saturation, a per-document transform, carries 61% of the gain and collection statistics 39%, and that IDF alone carries about a tenth, falling as the support budget grows.
3. A negative result for the pre-registered predictor, with its cause. Frequency diagnostics predict neither the margin nor the collection gain, because whether IDF and length help depends on relevance, which no statistic of the index measures.
4. A label-free predictor that works in retrieval. Agreement of each configuration with a reference ranking predicts all three components and the collection gain on held-out basis families and corpora. Five pre-registered tests of the mechanism pass on data the exploration did not use.

5. A transfer test outside retrieval, implemented. KV-page selection and expert routing adapters with native-parity checks are ready for the predictor to be frozen and applied.

4 Background

The correction needs three things of a representation, non-negative coordinates, a meaningful per-feature collection frequency df_j , and a magnitude that admits a length. Dense embeddings lack the first two, since every coordinate is signed and active in every document, so every df_j equals N . Sparse codes built from those embeddings have all three, and so does a selection count over KV pages or experts.

For a fixed linear map C and diagonal weights D , weighted feature scoring is bilinear in the original vectors, $S(q, d) = (Cq)^\top D(Cd) = q^\top C^\top DCd$, and it preserves every ranking of the original inner product when $C^\top DC = \alpha I$ with $\alpha > 0$. Split rectification, support truncation, saturation and a document-dependent length factor are not linear, so none of the representations here meets the condition, and the grid is needed to learn what the correction does. The identity, the sufficient conditions and a counterexample in which a diagonal rescaling reverses a ranking are checked numerically in the code.

5 Related Work

Term weighting. IDF was proposed as a statistical account of term specificity (Spärck Jones, 1972). Saturation came from approximating the two-Poisson model of within-document frequency (Robertson and Walker, 1994), and length normalisation was pivoted so that long documents are neither favoured nor penalised (Singhal et al., 1996). BM25 combines the three (Robertson and Zaragoza, 2009). Each part was motivated by word statistics, which is why it is open whether it helps features that are not words.

Retrieval scorers. Dense retrievers score by the inner product of embeddings (Karpukhin et al., 2020); the encoder used here is multilingual E5 (Wang et al., 2024b). Learned sparse retrieval keeps the inverted index and learns term weights, through contextual exact match (Gao et al., 2021) or expansion over the subword vocabulary in SPLADE (Formal et al., 2021b,a; Lassance et al., 2024). Lin and Ma (2021) and Nguyen et al. (2023) place these methods in one framework, and

Mackenzie et al. (2021) show that learned weights change the cost of query processing, which is why we report posting visits and index size. All of them score by inner product.

Learned concept vocabularies. Sparse autoencoders decompose dense activations into many non-negative features (Huben et al., 2024), and Top- K SAEs fix the number active per input (Gao et al., 2025). In retrieval, CL-SR indexes SAE-decomposed embeddings as concept units and weights them by IDF, and dropping that weighting costs 11 to 18 percent of its nDCG@10 on TREC-DL (Park et al., 2025), the one published ablation of the correction. SPLARE trains an SAE-based sparse retriever (Formal et al., 2026) and SAE-SPLADE replaces SPLADE’s language-model head with an SAE (Zong et al., 2026); both score by inner product. Quantised codes are the other route to a discrete vocabulary, through product quantisation (Jégou et al., 2011), residual quantisation (Chen et al., 2010) and their GPU implementations (Johnson et al., 2021).

BM25 on learned features. Clavié et al. (2026) show that SAE features of a frozen dense retriever have roughly Zipfian collection statistics and can be scored by BM25 directly. In their Appendix C, BM25 reaches 0.582 nDCG@10 against 0.381 for the inner product on SAE latent terms and 0.455 against 0.568 on SPLADE-v3, so the scorers swap places when the vocabulary changes. Han et al. (2026) find the two within about two points of R@1 on image retrieval and attribute the parity to top- k filtering already suppressing what IDF would down-weight. Raising their per-patch budget from 16 to 128 makes document frequencies converge and R@1 collapse from 0.508 to 0.038. Our budget sweep shows the matching trend, the share of the gain due to IDF falling as the budget grows (Section 7.4).

Predicting effectiveness without labels. Query performance prediction estimates retrieval quality from the query and the ranking alone (Carmel and Yom-Tov, 2010), through the divergence of the retrieved language model from the collection (Cronen-Townsend et al., 2002) or the spread of the top retrieval scores (Shtok et al., 2012). Retrieval systems can also be ranked without judgements by comparing each system with pseudo-relevant documents drawn from the pooled output (Soboroff et al., 2001). Our label-free predictor uses both

ideas, score-shape statistics and agreement with a reference ranking, to predict a difference between scorers, which neither line of work targets.

Attributing an effect to components. The Shapley value assigns a joint outcome to players by averaging marginal contributions over orderings (Shapley, 1953), and SHAP applies it to model predictions (Lundberg and Lee, 2017). We apply it to a complete 2^3 factorial, where it is exact and needs no sampling.

Selection inside the transformer. Quest scores KV pages by an inner product against page min-max key summaries (Tang et al., 2024). H2O keeps recent and heavy-hitter tokens (Zhang et al., 2023b), SnapKV keeps the positions an observation window attends to (Li et al., 2024), StreamingLLM keeps the first keys as attention sinks (Xiao et al., 2024), and Expected Attention prunes pairs that future queries are predicted to ignore (Devoto et al., 2025). Massive activations single out a few positions by magnitude (Sun et al., 2024). None of these uses the rate at which an entry is selected across queries. Long-context quality is measured by RULER (Hsieh et al., 2024).

Expert routing. Sparse mixture-of-experts layers route each token to its top- k experts (Shazeer et al., 2017; Lepikhin et al., 2021; Fedus et al., 2022). Balance is enforced by an auxiliary loss (Zoph et al., 2022) or, without one, by a load-driven bias (Wang et al., 2024a). DeepSeekMoE splits experts more finely and adds shared experts to sharpen their specialisation (Dai et al., 2024). Su et al. (2026) find a handful of experts by activation magnitude whose removal degrades the model badly. Our routing setting uses OLMoE (Muennighoff et al., 2025), which activates 8 of 64 experts per layer. To our knowledge no work treats retrieval scoring, KV selection and routing as one scoring problem.

Viability. Two papers report the correction winning on SAE features and one reports parity, all without a systematic vocabulary variation or a component attribution. The correction is cheap to switch on, sparse codes are cheap to build from one encoder, and label-free proxies for quality already work for single systems, so closing the gap needs no new model.

6 Method

Representations. All representations come from multilingual-e5-base (Wang et al., 2024b), frozen at a pinned revision. Seven bases are fitted on the same 200,000 MS MARCO passages (Bajaj et al., 2016) and nowhere else, namely split-rectified identity, Gaussian projection (768) and PCA (384), soft k -means over 4,096 centroids, product and residual quantisation (8×256), and a Top- K SAE with 8,192 features of which at most 32 are active. Each is applied at pooled granularity, to the mean of the content token states, and at token granularity, to each state before summing, giving fourteen representations. A secondary SAE shares one dictionary across both granularities to separate aggregation from fitting. SPLADE has its own backbone, so it is an external comparison outside the grid.

Corpus statistics. Supports are capped before any statistic, at 32 query and 128 document features in the primary budget. Document frequency, lengths and IDF are computed on each target corpus from the capped vectors, and retrieval is exact over the whole corpus. The primary grid uses $k_1 = 1.2$, $b = 0.75$ and seed zero.

Collections. The thirteen collections, from eleven corpora, are MS MARCO dev with TREC-DL 2019 and 2020 over the same documents, six BEIR sets (Thakur et al., 2021), LIMIT (Weller et al., 2026), and the Hindi, Bengali and Telugu portions of MIRACL (Zhang et al., 2023a), each evaluated within its language. LIMIT is built so that the top- k sets its queries need exceed what a low-dimensional embedding can represent.

Experiment families. The primary family (Table 1) was run twice, on a four-GPU cluster node for all thirteen collections and on a single-GPU workstation for the ten outside MS MARCO, which also ran every sensitivity family.

Significance. Comparisons use a paired bootstrap over queries that recomputes numerator and denominator, with Holm correction (Holm, 1979) within each pre-declared family, the all-on against all-off comparison and the three component main effects. The cluster run drew 2,000 samples, which puts the smallest attainable Holm p -value over 182 comparisons at 0.091, so significance is reported from the workstation run, which drew 10,000.

Family	Rows
primary (14 representations $\times$ 13 collections)	182
support budget 16/64, 32/128, 64/256, native	462
$k_1 \in \{0.6, 1.2, 2.4\}$, $b \in \{0, 0.5, 0.75\}$	630
basis seeds 1 and 2	100
shuffled and uniform df controls	84
shared SAE dictionary	20
union on the ten non-MS MARCO collections	1,226

Table 1: Retrieval rows, each with eight scoring configurations. The primary count is the cluster run’s; the other families and the union are the workstation run’s. Families share their default cells, so the union is smaller than the sum. The MS MARCO budget rows are pending (Section 8).

Diagnostics. None of the features describing a row reads relevance labels. The six pre-registered frequency diagnostics are the active fraction, the Zipf slope of the feature-frequency curve, the Gini coefficient of activation mass, frequency and activation mass in the most frequent 1% of features, and IDF variance; the primary ridge model uses the slope, IDF variance and frequent activation share. Index statistics describe the stored values, chiefly cv_x , the coefficient of variation of the stored document values, with the peak-to-mean ratio and the spread of document lengths. Ranking features are read from the stored top-100 list $R_c(q)$ of every configuration c and query q . The disruption of switch s is $1 - |R_c^{10}(q) \cap R_{c'}^{10}(q)|/10$ averaged over queries and over the four pairs (c, c') that differ only in s . Score-shape statistics of the kind used in query performance prediction (Shtok et al., 2012) are the coefficient of variation of the top-100 scores and the relative drop from the first to the tenth. The pseudo-quality $\tilde{Q}_c$ is the mean nDCG@10 of $R_c(q)$ when the top 10 of a reference retriever on the same query is taken as relevant, with binary gains. The references are text BM25, the dense encoder, and the two fused by reciprocal rank ($k = 60$ over their top 100). Replacing Q_c by $\tilde{Q}_c$ in Eq. (2) and in G gives a pseudo-label estimate of every target, $\tilde{\phi}_s$ and $\tilde{G}$, with no fitted parameter.

Validation. Predictors are ridge regressions with the scaler and the regularisation chosen inside training folds, and gradient-boosted trees with fixed settings. Folds hold out a whole basis family (linear, clustered, quantised, learned) or a whole corpus, reported separately, since a new basis and a new corpus are different claims. A fit on primary rows is also scored on the budget, seed and parameter

Collection	BM25	E5	000	111	best 111
MS MARCO dev	.222	.425	.013	.130	.318
TREC-DL 19	.456	.704	.036	.255	.604
TREC-DL 20	.484	.697	.035	.251	.556
ArguAna	.411	.445	.021	.130	.371
FiQA	.236	.382	.013	.085	.266
NFCorpus	.307	.325	.064	.115	.285
SCIDOCs	.152	.172	.009	.044	.120
SciFact	.662	.694	.089	.197	.419
TREC-COVID	.582	.695	.134	.246	.508
LIMIT	.939	.016	.001	.004	.032
MIRACL hi	.357	.583	.011	.124	.328
MIRACL bn	.407	.701	.025	.216	.490
MIRACL te	.331	.751	.019	.259	.579

Table 2: nDCG@10 of text BM25, the dense encoder, and the fourteen sparse representations with the plain inner product (000) and full correction (111), averaged, with the best representation’s 111. The best is the token-level SAE everywhere except LIMIT (token k -means).

rows, which share collections with it.

Transfer adapters. The attention adapter implements Quest-style page criticality over sixteen-token pages with dense prefill and sparse decode, counts how often each page enters the top- k over causal prefill positions, and applies Eq. (1) with pages for documents; all-off calls the native selector, and an H2O reference shares the budget accounting. The routing adapter applies the same correction to OLMoE router probabilities for selection while combining with the original gate weights; all-off calls the native router. Six tests confirm exact native parity, causal calibration, budget charging including sink pages, and the gate mixture.

7 Progress

7.1 The pipeline reproduces itself

The two runs used the same imported bases and pinned data. Over the 140 primary rows they share, the median absolute difference in all-on nDCG@10 is 0.0007 (maximum 0.0077, from fp16 rounding on different GPUs), and the margin sign agrees in 136 of the 137 rows where it is defined. The text BM25 baselines agree to seven decimals. Basis seeds move all-on nDCG@10 by a median standard deviation of 0.006 over 50 representation-collection groups. The conceptual replication of the Appendix C reversal of Clavié et al. (2026) needs SPLADE-v3 and is scheduled (Section 8).

7.2 Retrieval quality

In the primary grid (Table 2), the plain inner product of the capped sparse codes nearly fails, with

Rows	IDF	Sat.	Len.	IDF + len.
cluster, 182 primary	.10	.62	.28	.38 (.36–.41)
workstation, 140 primary	.10	.61	.29	.39 (.36–.42)
budget 16/64	.16	.60	.25	.40
budget 64/256	.08	.57	.35	.43
native, 3 collections	−.02	.55	.47	.45

Table 3: Share of the all-on minus all-off gain assigned to each switch by Eq. (2). Intervals resample whole collections. The primary budget is 32/128; the other budget rows cover the same ten collections as the workstation primary row, and native support covers ArguAna, NFCorpus and SciFact.

mean nDCG@10 below 0.14 on every collection, and full correction raises the mean by a factor of between 1.8 and 14. Token granularity is far stronger than pooled (mean all-on 0.219 against 0.097), and the token-level SAE is the best representation on twelve collections, reaching 0.60 on TREC-DL 2019 against 0.70 for the dense encoder it was built from. On the ten workstation collections the pooled SAE reaches 0.095 with its own dictionary and 0.291 with the token-trained one, against 0.339 at token level, so most of the pooled SAE’s deficit comes from fitting its dictionary on pooled vectors, and aggregation order accounts for the rest. LIMIT behaves as its construction predicts, BM25 on words reaching 0.94 where the dense encoder and every representation built from it stay near zero.

7.3 Whether the correction helps

The margin is positive in 90% of the 179 primary rows with a defined margin. Over the 140 rows of the workstation run, Holm-corrected all-on against all-off comparisons are significant in 99, and the main effects of saturation, length and IDF in 93, 79 and 56, with mean effects of +0.065, +0.032 and +0.013 nDCG@10. Of the 182 primary rows, full correction is the best of the eight configurations in only 43; the most frequent winner, in 51, is saturation with length normalisation and no IDF. On the workstation run the all-off configuration is below 0.01 nDCG@10 in 42% of rows, and the margin falls between 1.5 and 3.5 in 83%, so the margin mostly records that the uncorrected scorer is broken and leaves little variation to predict.

7.4 Where the gain comes from

Saturation carries about 61% of the gain and IDF about 10% on both runs (Table 3). Once satu-

Basis	Gran.	111	cv_x	ϕ_{sat}	IDF + len.	$G > 0$
identity	pooled	.147	0.37	.043	.68	1.00
	token	.142	0.60	.060	.55	.77
Gaussian	pooled	.098	0.29	.020	.77	.85
	token	.088	0.55	.029	.64	.92
PCA	pooled	.222	0.75	.138	.24	.69
	token	.311	1.35	.155	.45	.46
k -means	pooled	.018	0.04	.000	.94	.31
	token	.185	1.71	.125	.11	.85
PQ	pooled	.012	0.00	.000	1.00	.62
	token	.144	1.44	.099	.09	.69
RQ	pooled	.056	0.00	.000	1.00	.92
	token	.292	2.10	.176	.18	.69
SAE	pooled	.127	0.18	.014	.66	.77
	token	.373	3.31	.193	.32	.46

Table 4: Per representation, mean all-on nDCG@10, mean saturation Shapley value, share of the gain from IDF and length, and the fraction of collections where collection statistics help after saturation ($G > 0$), over the 13 primary collections, with the mean cv_x of the stored values over the ten whose indexes were measured.

ration is applied, collection statistics add 0.015 nDCG@10 on average on the workstation run and 0.021 on the cluster run, and help in about two thirds of rows. On the ten workstation collections, the share of IDF falls from 16% at the 16/64 budget to 8% at 64/256 while that of length rises from 25% to 35%, and at native support on three collections IDF’s share is below zero. Han et al. (2026) report the same trend for image features and attribute it to document frequencies converging as budgets grow. On the five collections of the parameter family, all-on nDCG@10 at $b = 0$ falls as k_1 grows (0.170, 0.164, 0.153), and at $b = 0.75$ it is flat (0.175 to 0.176), so weaker saturation costs most when length normalisation is off.

Replacing document frequencies by shuffled ones on three collections turns the mean IDF Shapley value from +0.011 to −0.011 (real larger in 83% of 42 rows, one-sided Wilcoxon $p = 8.9 \times 10^{-7}$); uniform frequencies remove it by construction, and do not lower all-on nDCG@10 (0.157 against 0.147, $p = 0.15$). Real frequencies carry information, but little of it shows in nDCG@10.

Across representations (Table 4), saturation’s contribution rises with cv_x , and the extreme case is exact. Pooled PQ and RQ codes store equal values with equal lengths ($cv_x = 0$), and their Shapley values for saturation and length are exactly zero in all 26 of their cluster rows and all 166 workstation rows, as the mechanism below requires. The share of the gain from collection statistics is largest for the weakest representations. For the two strongest,

Target	Features	Family	Corpus	Params
ϕ_{sat}	pre-registered	-.46	.14	.02
	index	.47	.42	.50
	pseudo-labels	.56	.62	.76
ϕ_{len}	pre-registered	-.30	.14	.00
	index	-.16	.25	-.02
	pseudo-labels	.49	.70	.77
ϕ_{idf}	pre-registered	-.01	.00	.02
	index	.02	.02	.07
	pseudo-labels	.40	.21	.73
G	pre-registered	-.07	-.02	.16
	index	-.05	-.09	.15
	pseudo-labels	.38	.52	.62
	pseudo-labels, E5	.51	.66	.80

Table 5: Held-out R^2 of ridge regression fitted on 139 primary rows, holding out basis families, corpora, or predicting the 560 rows at other k_1 and b . Each tier adds features to the one above it. Pseudo-labels use text BM25 unless marked E5.

token-level SAE and PCA, collection statistics help after saturation on fewer than half the collections, which is where predicting the decision matters.

7.5 The pre-registered predictor

The primary ridge model predicts the margin with held-out R^2 of 0.04 across basis families and 0.02 across corpora on the cluster run, and -0.11 and 0.14 on the workstation run with the rows that respond to fewer than two switches excluded. Its sign accuracy equals the majority-sign baseline in every case (0.899 and 0.924). The six-diagnostic model reaches $R^2 = 0.30$ across corpora on the workstation run but -0.54 across basis families, and its sign accuracy is never more than 0.03 above the majority sign. By the pre-registered criterion the frequency diagnostics carry no usable information about the margin.

7.6 What predicts the gain

Mechanism. With length normalisation off, saturation maps a stored value f to $g(f) = (k_1 + 1)f/(f + k_1)$, and $g(f)/f$ falls as f grows, so it shrinks large values relative to small ones. If every stored value were equal, g would rescale all scores alike and change no ranking; the change grows with how unequal the values are, which cv_x measures, and a smaller k_1 bends g more. How much saturation changes the ranking is a property of the index, and the change helps in 91% of cluster primary rows outside the two equal-valued codes. Length normalisation and IDF change the ranking by an amount the index fixes, but whether the change helps depends on whether heavy documents

are over-rewarded relative to their relevance, and whether rare features separate relevant documents from the rest. Both are properties of the index jointly with relevance, so no statistic of the index can fix their sign.

Prediction. Index statistics predict saturation and fail for IDF and the collection gain, and the frequency diagnostics predict nothing (Table 5). Pseudo-labels carry the relevance information the index lacks and predict every target. The Shapley values and G are fixed linear combinations of the eight Q_c , so a pseudo-quality that orders configurations like the true one gives correlated targets, and it does, with a median within-row Spearman correlation of 0.98 between pseudo and true nDCG@10 over the eight configurations of 1,112 rows. Used with no fitting, the pseudo-label estimates rank the true values with Spearman 0.91 for saturation, 0.82 for length, 0.83 for IDF and 0.66 for G on the primary rows. Gradient-boosted trees do no better on the family and corpus axes.

Decision. The decision rule uses collection statistics only where the unfitted pseudo-label estimate $\tilde{G}$ is positive, and has no fitted parameter. It raises mean nDCG@10 on the primary rows from 0.1429 (always) to 0.1478, against 0.1510 for an oracle, recovering 61% of the headroom with a collection-resampled 95% interval of 0.0003 to 0.0101 above always. The rule recovers 67% at other budgets and 65% at other parameters, and 22% at other seeds, where the interval reaches zero. Across all rows, the rule gains most on SciFact, where always using collection statistics loses 0.012 on average, and on NFCorpus, where always using them gains nothing; it is at chance only on LIMIT and TREC-COVID, where the true gain is near zero.

Choice of reference. Agreement with the dense reference partly measures how much of the embedding a configuration keeps, since the sparse codes are built from the same encoder. Agreement with BM25 may be favoured by the judgements, since lexical systems contributed to the pools behind several collections. The result holds with either (the E5 row of Table 5), within-row agreement is 0.98 for both, and the unfitted rule with the dense reference recovers 45 to 75% of the headroom across the four groups of rows. All of this analysis was chosen after the pre-registered result and is exploratory.

Week	Dates	Work	Owner
1	3–9 Oct	Resume the MS MARCO budget sweep (84 rows) after the cluster maintenance. Run SPLADE-v3 as the external comparison and a conceptual replication of the Appendix C reversal. Commit the second pre-registration, the pseudo-label rule tested on the unseen MS MARCO budget rows, and the frozen transfer predictors. Validation runs of both adapters on the cluster, native parity at scale, calibration cost and memory.	Gupta, Singh Sankar, Tripathy
2	10–16 Oct	Setting B primary grid on Llama-3.2-1B-Instruct, all 13 RULER tasks with 500 examples, 4K and 8K contexts, budgets of 512 and 1,024 tokens, with H2O. Second confirmatory round in retrieval.	Sankar, Singh
3	17–23 Oct	Setting C on OLMoE-1B-7B, WikiText-2 negative log-likelihood and zero-shot MMLU, with the additive loss-free-balancing analogue. Written proof and checks of the bilinear identity and its ranking conditions.	Tripathy, Singh
4	24–28 Oct	Transfer scoring with uncertainty for both predictors, per setting. Figures and tables regenerated from saved artifacts.	all
5	29–31 Oct	Final report, slides, code archive.	all

Table 6: Schedule to the final submission. If time runs short we cut, in order, GSM8K and expert preservation, the 8K context in Setting B, MMLU in Setting C (keeping WikiText), and the Appendix C replication.

Hypothesis	Result
H1 IDF + length share at default in [.36, .42]	.393
H2 ϕ_{sat} larger at $k_1 = 0.6$ than 2.4	.093 vs .080, $p = 3.6 \times 10^{-5}$
H3 within-corpus Spearman of cv_x with $\phi_{\text{sat}} \geq 0.3$	.89 and .89
H4 primary-fitted cv_x model ranks ϕ_{sat} at $\rho \geq 0.5$	.87 and .90
H5 $\phi_{\text{len}} = 0$ exactly at $b = 0$	max 0.0, 210 rows
H6 ϕ_{len} larger at $b = 0.75$ than 0.5	.045 vs .029, $p = 3.6 \times 10^{-5}$

Table 7: Confirmatory tests. All pass. H3 and H4 are tested at $k_1 = 0.6$ and 2.4, compression strengths the model was not fitted on.

7.7 Confirmatory tests

Six hypotheses were committed to the repository before their tests ran, on the k_1 and b family, whose 630 rows the exploration up to that point had not used. H2 to H6 are Holm-corrected together; H1 overlaps the exploratory data and checks consistency only.

All six pass (Table 7). H5 is a check of the implementation. H2, H3, H4 and H6 confirm that the switches behave as their mechanisms predict when their strength is varied, and that the value-dispersion predictor of saturation keeps its ranking at settings it was not fitted on.

7.8 Transfer settings

The adapters and their parity tests are complete (Section 6); no production run has been made. The predictor and the acceptance thresholds will be frozen before any corrected transfer outcome is

seen.

8 Remaining Plan

The pre-registered predictor is kept and will be carried to attention and routing as registered. The pseudo-label predictor is carried beside it, with a reference native to each setting, exact attention mass for KV pages, and the dense softmax mixture over all experts for routing, both label-free and computed from the uncorrected model. Before any transfer outcome is seen, both predictors, the references and the thresholds are committed, in the first week of the schedule in Table 6.

9 Limitations

All representations come from one encoder, so basis and encoder effects are not separable, and E5 was trained with MIRACL data, so the Indic results do not concern unseen languages. The sparse codes are far below the dense encoder they come from, which bounds how far the attribution speaks to strong learned sparse retrievers; the SPLADE comparison is meant to test that. The label-free predictor needs queries and a second retriever run, and its evidence is exploratory until the second pre-registered round. Each reference carries a bias, and the result holding with both makes a single bias an unlikely explanation. Routing is inference-time only, so nothing is claimed about training dynamics.

References

- Payal Bajaj, Daniel Campos, Nick Craswell, Li Deng, Jianfeng Gao, Xiaodong Liu, Rangan Majumder, Andrew McNamara, Bhaskar Mitra, Tri Nguyen, Mir Rosenberg, Xia Song, Alina Stoica, Saurabh Tiwary, and Tong Wang. 2016. MS MARCO: A human generated machine reading comprehension dataset. *arXiv preprint arXiv:1611.09268*.
- David Carmel and Elad Yom-Tov. 2010. *Estimating the Query Difficulty for Information Retrieval*. Synthesis Lectures on Information Concepts, Retrieval, and Services. Morgan & Claypool.
- Yongjian Chen, Tao Guan, and Cheng Wang. 2010. Approximate nearest neighbor search by residual vector quantization. *Sensors*, 10(12):11259–11273.
- Benjamin Clavié, Sean Lee, Aamir Shakir, and Makoto P. Kato. 2026. Latent terms: Dense retrievers contain trivially extractable BM25-ready Zipfian vocabularies. *arXiv preprint arXiv:2605.29384*.
- Steve Cronen-Townsend, Yun Zhou, and W. Bruce Croft. 2002. Predicting query performance. In *Proceedings of the 25th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 299–306. ACM.
- Damai Dai, Chengqi Deng, Chenggang Zhao, R. X. Xu, Huazuo Gao, Deli Chen, Jiashi Li, Wangding Zeng, Xingkai Yu, Y. Wu, Zhenda Xie, Y. K. Li, Panpan Huang, Fuli Luo, Chong Ruan, Zhifang Sui, and Wenfeng Liang. 2024. DeepSeekMoE: Towards ultimate expert specialization in mixture-of-experts language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1280–1297, Bangkok, Thailand. Association for Computational Linguistics.
- Alessio Devoto, Maximilian Jeblick, and Simon Jégou. 2025. Expected attention: KV cache compression by estimating attention from future queries distribution. *arXiv preprint arXiv:2510.00636*.
- William Fedus, Barret Zoph, and Noam Shazeer. 2022. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *Journal of Machine Learning Research*, 23(120):1–39.
- Thibault Formal, Carlos Lassance, Benjamin Piwowarski, and Stéphane Clinchant. 2021a. SPLADE v2: Sparse lexical and expansion model for information retrieval. *arXiv preprint arXiv:2109.10086*.
- Thibault Formal, Maxime Louis, Hervé Déjean, and Stéphane Clinchant. 2026. Learning retrieval models with sparse autoencoders. In *International Conference on Learning Representations (ICLR)*.
- Thibault Formal, Benjamin Piwowarski, and Stéphane Clinchant. 2021b. SPLADE: Sparse lexical and expansion model for first stage ranking. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 2288–2292. ACM.
- Leo Gao, Tom Dupre la Tour, Henk Tillman, Gabriel Goh, Rajan Troll, Alec Radford, Ilya Sutskever, Jan Leike, and Jeffrey Wu. 2025. Scaling and evaluating sparse autoencoders. In *International Conference on Learning Representations (ICLR)*.
- Luyu Gao, Zhu Yun Dai, and Jamie Callan. 2021. COIL: Revisit exact lexical match in information retrieval with contextualized inverted list. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT)*, pages 3030–3042. Association for Computational Linguistics.
- Donghoon Han, Eunhwan Park, and Seunghyeon Seo. 2026. Visual words meet BM25: Sparse auto-encoder visual word scoring for image retrieval. *arXiv preprint arXiv:2603.05781*.
- Sture Holm. 1979. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70.
- Cheng-Ping Hsieh, Simeng Sun, Samuel Kriman, Shantanu Acharya, Dima Rekish, Fei Jia, and Boris Ginsburg. 2024. RULER: What’s the real context size of your long-context language models? In *Conference on Language Modeling (COLM)*.
- Robert Huben, Hoagy Cunningham, Logan Smith, Aidan Ewart, and Lee Sharkey. 2024. Sparse autoencoders find highly interpretable features in language models. In *International Conference on Learning Representations (ICLR)*.
- Hervé Jégou, Matthijs Douze, and Cordelia Schmid. 2011. Product quantization for nearest neighbor search. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 33(1):117–128.
- Jeff Johnson, Matthijs Douze, and Hervé Jégou. 2021. Billion-scale similarity search with GPUs. *IEEE Transactions on Big Data*, 7(3):535–547.
- Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. 2020. Dense passage retrieval for open-domain question answering. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 6769–6781, Online. Association for Computational Linguistics.
- Carlos Lassance, Hervé Déjean, Thibault Formal, and Stéphane Clinchant. 2024. SPLADE-v3: New baselines for SPLADE. *arXiv preprint arXiv:2403.06789*.
- Dmitry Lepikhin, Hyoungho Lee, Yuanzhong Xu, Dehao Chen, Orhan Firat, Yanping Huang, Maxim Krikun, Noam Shazeer, and Zhifeng Chen. 2021. GShard: Scaling giant models with conditional computation and automatic sharding. In *International Conference on Learning Representations (ICLR)*.

- Yuhong Li, Yingbing Huang, Bowen Yang, Bharat Venkitesh, Acyr Locatelli, Hanchen Ye, Tianle Cai, Patrick Lewis, and Deming Chen. 2024. SnapKV: LLM knows what you are looking for before generation. In *Advances in Neural Information Processing Systems 37 (NeurIPS)*.
- Jimmy Lin and Xueguang Ma. 2021. A few brief notes on DeepImpact, COIL, and a conceptual framework for information retrieval techniques. *arXiv preprint arXiv:2106.14807*.
- Scott M. Lundberg and Su-In Lee. 2017. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems 30 (NIPS)*.
- Joel Mackenzie, Andrew Trotman, and Jimmy Lin. 2021. Wacky weights in learned sparse representations and the revenge of score-at-a-time query evaluation. *arXiv preprint arXiv:2110.11540*.
- Niklas Muennighoff, Luca Soldaini, Dirk Groeneveld, Kyle Lo, Jacob Morrison, Sewon Min, Weijia Shi, Pete Walsh, Oyvind Tafjord, Nathan Lambert, Yuling Gu, Shane Arora, Akshita Bhagia, Dustin Schwenk, David Wadden, Alexander Wettig, Binyuan Hui, Tim Dettmers, Douwe Kiela, and 5 others. 2025. OLMoE: Open mixture-of-experts language models. In *International Conference on Learning Representations (ICLR)*.
- Thong Nguyen, Sean MacAvaney, and Andrew Yates. 2023. A unified framework for learned sparse retrieval. In *European Conference on Information Retrieval (ECIR)*, pages 101–116. Springer.
- Seongwan Park, Taeklim Kim, and Youngjoong Ko. 2025. Decoding dense embeddings: Sparse autoencoders for interpreting and discretizing dense retrieval. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 26468–26485, Suzhou, China. Association for Computational Linguistics.
- Stephen Robertson and Hugo Zaragoza. 2009. The probabilistic relevance framework: BM25 and beyond. *Foundations and Trends in Information Retrieval*, 3(4):333–389.
- Stephen E. Robertson and Steve Walker. 1994. Some simple effective approximations to the 2-Poisson model for probabilistic weighted retrieval. In *Proceedings of the 17th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '94)*, pages 232–241. Springer.
- Lloyd S. Shapley. 1953. A value for n -person games. In Harold W. Kuhn and Albert W. Tucker, editors, *Contributions to the Theory of Games, Volume II*, number 28 in Annals of Mathematics Studies, pages 307–317. Princeton University Press.
- Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarczyk, Andy Davis, Quoc V. Le, Geoffrey E. Hinton, and Jeff Dean. 2017. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. In *International Conference on Learning Representations (ICLR)*.
- Anna Shtok, Oren Kurland, David Carmel, Fiana Raiber, and Gad Markovits. 2012. Predicting query performance by query-drift estimation. *ACM Transactions on Information Systems*, 30(2):1–35.
- Amit Singhal, Chris Buckley, and Mandar Mitra. 1996. Pivoted document length normalization. In *Proceedings of the 19th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 21–29. ACM.
- Ian Soboroff, Charles Nicholas, and Patrick Cahan. 2001. Ranking retrieval systems without relevance judgments. In *Proceedings of the 24th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 66–73. ACM.
- Karen Spärck Jones. 1972. A statistical interpretation of term specificity and its application in retrieval. *Journal of Documentation*, 28(1):11–21.
- Zunhai Su, Qingyuan Li, Hao Zhang, Weihao Ye, Qibo Xue, Yulei Qian, Yuchen Xie, Ngai Wong, and Kehong Yuan. 2026. Unveiling super experts in mixture-of-experts large language models. In *International Conference on Learning Representations (ICLR)*.
- Mingjie Sun, Xinlei Chen, J. Zico Kolter, and Zhuang Liu. 2024. Massive activations in large language models. In *Conference on Language Modeling (COLM)*.
- Jiaming Tang, Yilong Zhao, Kan Zhu, Guangxuan Xiao, Baris Kasikci, and Song Han. 2024. Quest: Query-aware sparsity for efficient long-context LLM inference. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235, pages 47901–47911. PMLR.
- Nandan Thakur, Nils Reimers, Andreas Rücklé, Abhishek Srivastava, and Iryna Gurevych. 2021. BEIR: A heterogeneous benchmark for zero-shot evaluation of information retrieval models. In *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks (NeurIPS Datasets and Benchmarks)*.
- Lean Wang, Huazuo Gao, Chenggang Zhao, Xu Sun, and Damai Dai. 2024a. Auxiliary-loss-free load balancing strategy for mixture-of-experts. *arXiv preprint arXiv:2408.15664*.
- Liang Wang, Nan Yang, Xiaolong Huang, Linjun Yang, Rangan Majumder, and Furu Wei. 2024b. Multilingual E5 text embeddings: A technical report. *arXiv preprint arXiv:2402.05672*.
- Orion Weller, Michael Boratko, Iftexhar Naim, and Jinhyuk Lee. 2026. On the theoretical limitations of embedding-based retrieval. In *International Conference on Learning Representations (ICLR)*.

Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, and Mike Lewis. 2024. Efficient streaming language models with attention sinks. In *International Conference on Learning Representations (ICLR)*.

Xinyu Zhang, Nandan Thakur, Odunayo Ogundepo, Ehsan Kamalloo, David Alfonso-Hermelo, Xiaoguang Li, Qun Liu, Mehdi Rezagholizadeh, and Jimmy Lin. 2023a. MIRACL: A multilingual retrieval dataset covering 18 diverse languages. *Transactions of the Association for Computational Linguistics*, 11:1114–1131.

Zhenyu Zhang, Ying Sheng, Tianyi Zhou, Tianlong Chen, Lianmin Zheng, Ruisi Cai, Zhao Song, Yuandong Tian, Christopher Ré, Clark Barrett, Zhangyang Wang, and Beidi Chen. 2023b. H2O: Heavy-hitter oracle for efficient generative inference of large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, pages 34661–34710.

Yuxuan Zong, Mathias Vast, Basile Van Cooten, Laure Soulier, and Benjamin Piwowarski. 2026. [From tokens to concepts: Leveraging SAE for SPLADE](#). In *Proceedings of the 49th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '26)*, Melbourne, VIC, Australia. ACM.

Barret Zoph, Irwan Bello, Sameer Kumar, Nan Du, Yanping Huang, Jeff Dean, Noam Shazeer, and William Fedus. 2022. ST-MoE: Designing stable and transferable sparse expert models. *arXiv preprint arXiv:2202.08906*.